Fiscal Year:	FY 2009	**Task Last Updated:**	FY 11/03/2008
PI Name:	Azzam, Edouard Ph.D.		
Project Title:	The Role of Gap-Junction Communication and Oxidative Metabolism in the Biological Effects of Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	azzamei@njms.rutgers.edu	**Fax:**	FY (973) 972-1865
PI Organization Type:	UNIVERSITY	**Phone:**	(973) 972-5323
Organization Name:	RUTGERS Biomedical and Health Sciences - New Jersey Medical School		
PI Address 1:	New Jersey Medical School Cancer Center - Department of Radiology		
PI Address 2:	205 S Orange Ave, Cancer center - F1012		
PI Web Page:			
City:	Newark	**State:**	NJ
Zip Code:	07103	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Toledo , Sonia (UMDNJ-New Jersey Medical School) Harris, Andrew (New Jersey Medical School (UMDNJ)) Howell, Roger (University of Medicine and Dentistry of New Jersey) Pain, Debkumar (UMDNJ - New Jersey Medical School)		
Grant/Contract No.:	NNJ06HD91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this project is to investigate biochemical and molecular aspects of two inter-related fundamental biological processes, namely gap-junction intercellular communication and oxidative metabolism, in determining human cellular responses to exposures of very low to moderate fluences of protons (of energies such as found in the space environment) or high charge (Z) and high energy (E) particles (1 GeV/nucleon 56Fe ions). Gap junctions are found between most cells in the body, and the molecular signaling they mediate is crucial for normal development, physiology and response to disease. By allowing direct intercellular transfer of cytoplasmic molecules, they provide a powerful pathway for direct molecular signaling between cells. On the other hand, oxidative metabolism is primarily regulated by mitochondria, which mediate respiration and ATP synthesis in eukaryotic cells. They also participate in numerous indispensable metabolic pathways. We hypothesize that exposure of human cells to high or low fluences of protons and/or HZE particles results in distinct molecular and biochemical changes not only in irradiated but also neighboring bystander cells. These events are modulated by oxidative metabolism and gap-junction coupling, which together induce persistent changes in mitochondria and in nuclear DNA in progeny of irradiated and affected bystander cells. In Aim 1, we propose to define the aspect(s) of junctional communication leading to proton- and high charge/high energy (HZE) particle-induced toxic effects. Specifically, we will examine the role of different connexin proteins in propagating death-inducing (growth delay, apoptosis) or pro-survival effects (proliferative capacity) among irradiated cells and also between irradiated and non-irradiated cells. Aim 2 is to identify specific connexin channel permeabilities that correlate with intercellular propagation of toxic or protective effects in cultures exposed to HZE particles or protons. An emphasis will be to determine the nature of molecules communicated among irradiated cells and between irradiated and non-irradiated cells. In Aim 3, we will evaluate mitochondrial function in cells exposed to low fluences of high energy charged particles. Specifically, we will examine mitochondrial protein import, mitochondrial protein synthesis, assembly of large protein complexes including respiratory complexes, and oxidative damage to mitochondrial proteins and DNA. We will also evaluate the persistence of oxidative effects in the progeny of exposed cells and correlate the observed results with the overall cellular redox environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed experiments relate to Risk No. 28, subsection 28b (molecular studies on carcinogenesis) in NASA's 'Roadmap'. In addition to revealing signaling pathways underlying non-targeted effects of ionizing radiation, the outcome of this proposal may provide insight into space radiation-induced degenerative diseases such as cognitive defects and cardiovascular disorders (Risks No. 29 and 30). It will provide a mechanistic basis for understanding exposures and risks to human from low fluences of high and low linear energy transfer radiations. By identifying the nature of molecules communicated among irradiated and neighboring non-irradiated cells, it may contribute to remediation strategies against harmful effects of space radiation, and is also relevant to heavy ion and proton radiotherapy.
	Repeat experiments, during five NSRL runs, with 1 GeV/amu iron ions or 1 GeV protons have revealed consistent data on the following: I. The Role of Intercellular Communication in Space Radiation Effects (related to Specific Aims 1 and 2): a) Using a layered tissue culture strategy that allows separation of irradiated and bystander cells, our experiments have shown that stressful effects are propagated from energetic iron ion (1 GeV/amu)-exposed AG1522 normal human fibroblasts to neighboring non-irradiated quiescent AG1522 cells. Micronuclei, a surrogate form of DNA damage, were induced in G0/G1-phase bystander cells co-cultured with cells from cultures exposed to mean doses of 200 cGy or 25 cGy. Whereas at 200 cGy every cell in the exposed population is traversed by multiple iron ion tracks, at 25 cGy, a large proportion of cells (~30%) are traversed by only one ion track. Gap-junction intercellular communication (GJIC) is the major mediator of the induced bystander effect detected in our experimental system. Collectively, our data are consistent with the concept that the linear energy transfer of the radiation (LET) is an important factor that determines the radiation-induced propagation of stressful events leading to DNA damage in bystander cells. In agreement with the latter concept, micronuclei were not induced in bystander cells co-cultured with proton-irradiated (200 cGy, 1 GeV) cells. Together, these results are consistent with our observation of significant stressful bystander effects expressed after exposure of confluent cell populations to very low doses (0.1-1 cGy) from 3.2 MeV a-particles (a high LET radiation), and the lack of propagation of such stressful effects from cultures exposed to low or high dose Cs-137 gamma rays (a low LET radiation). b) Our observation that stressful bystander effects are induced following exposure to high charge/high energy (HZE) particles is highly significant to the studies proposed in Aim 2 of our proposal. These studies, which are in progress, particularly seek to identify specific channel permeabilities that correlate with intercellular propagation of toxic or protective effects in cell cultures exposed to HZE particles or protons. c) Consistent with the participation of gap-junction intercellular communication in the cellular response to space radiation, exposure of confluent cell cultures to energetic iron ions resulted in up-regulation of connexin 43 (Cx43) protein, and irradiated cells were capable of functional communication via gap-junctions with each other, and with non-irradiated bystander cells. Significantly in AG1522 cell cultures exposed to mean doses as low as 10 cGy from 1 GeV/amu iron ions, the hyperphosphorylated form of Cx43 and caveolin-1 are preferentially located in cholesterol-rich domains (lipid rafts) of the plasma membrane. Analyses are in progress to examine the expression of Cx43 and caveolin-1 in plasma membranes of cells from cultures exposed to a mean dose of 1 cGy from 1 GeV/amu iron ions. d) Further support of direct intercellular communication involvement in cellular responses to space radiation was derived from confluent or sparse human fibroblast cultures exposed to 1 GeV/amu iron ions. Induction of micronuclei in cells derived from confluent cultures, whereby contiguous cells communicate via junctional channels, was significantly greater than in cells derived from sub-confluent cultures. e) In experiments designed to investigate aspects of gap-junction intercellular communication leading to propagation of energetic proton or high charge/high energy particle (HZE)-induced biological effects, we have shown that junctional channel permeability is important in determining the sensitivity of human cells to space radiation. Whereas Cx26 sensitized HeLa cells to 1 GeV/amu iron ions or 1GeV protons, Cx32 protected cells against the lethal effects of these radiations. Similar protective and sensitizing effects of Cx32 and Cx26, respectively, were obtained in cell cultures exposed to 3.2 MeV a-particles. f) In an effort to further understand the biological effects of space radiation, we have compared the cell killing potential

Task Progress:	of 1 GeV protons (LET ~0.2 keV/µm) or 1 GeV/amu iron ions (LET ~151 keV/µm) in normal human diploid fibroblasts (AG1522), and compared the results with data obtained in AG1522 cells exposed, in parallel, to Cs-137 gamma-rays (LET~0.9 keV/µm) or 3.5 MeV a-particles (LET ~113 keV/µm). As expected, a higher level of cell killing and DNA damage was induced in confluent cultures (98% in G0/G1) exposed to a-particles or energetic iron ions than energetic protons or gamma-rays. At 10% survival level, the relative biological effectiveness (RBE) of energetic protons, a-particles and iron ions compared to gamma-rays were ~ 1.02, 5.8 and 3.2, respectively. Significant potentially lethal damage recovery (PLDR) occurred when cells were incubated for 3 h after exposure to gamma-rays or energetic protons. In contrast, no or minimal PLDR occurred in a-particle- or iron ion-irradiated cells, respectively. Consistent with these observations, greater level of DNA damage was expressed during the post-incubation period following the high LET exposures. Support for gap-junction intercellular communication in propagating stressful effects among irradiated cells, which results in an increase in DNA damage during the incubation period post-high LET irradiation was derived from experiments with gap junction inhibitors. The increase in DNA damage expressed during the incubation period post a-particle or iron ion irradiation was inhibited when cells were treated with a-glycyrrhetinic acid (AGA), a gap-junction inhibitor. II. The Role of Oxidative Metabolism (Specific Aim 3) g) Our repeated experiments have now shown that low and high mean doses of 1GeV/amu iron ions disrupt mitochondrial physiology. Exposure of confluent, density inhibited normal human fibroblast cultures (AG1522) to 10 or 100 cGy resulted in similar impairment of mitochondrial protein import capacity. The decrease in mitochondrial import in exposed cultures correlated with modulation of mitochondrial membrane potential. With particular relevance to bystander effects, at 100 cGy every cell in the exposed population is traversed by multiple radiation tracks; in contrast, at 10 cGy, only 50% of cells would be traversed through the nucleus by an irradiating iron ion. Though, the non-irradiated cells may be traversed by delta-rays (however, in the protocol used, this is expected to be minimal). Thus, these results further support the expression of bystander effects in cell cultures exposed to HZE particles. The impairment of protein import by low fluence energetic iron particles is indicative of persistent oxidative stress in the HZE-exposed cell populations. h) The persistence of oxidative stress in cells exposed to space radiation is supported by analyses of aconitase activity in control and irradiated AG1522 cells. Exposure of AG1522 cell cultures to doses as low as 10 cGy from 1 GeV/amu protons or iron ions down-regulate mitochondrial aconitase (ACO2) activity. These data support the strong connection between oxidative stress and damages that follow radiation exposure. i) Oxidative stress and DNA damage in progeny of irradiated and bystander cells from cultures exposed to space radiation. Compared to respective control cells, stressful effects persist in the progeny of AG1522 normal human fibroblasts exposed to low and high dose from 1GeV/amu iron ions. The elevated levels of micronucleus formation may be observed in progeny cells up to 20 population doublings after exposure. The increase in micronucleus formation in progeny of iron ion-irradiated (1 GeV/amu) cells correlated with significant increases in mitochondrial superoxide anions and amount of cellular mitochondria in progeny cells, as revealed by the specific dyes MitoSOX and Mitotracker green, respectively. In contrast to data with energetic iron ions, DNA damage and oxidative stress did not persist in progeny of proton-irradiated (1 GeV) cells. III. Response of the progeny of bystander cells to a challenge dose of ionizing radiation (related to Specific Aim 3): j) Preliminary experiments have shown that the progeny of bystander cells derived from HZE-particle irradiated cultures express similar responses, as control cells, to a new challenge of ionizing radiation. Exposure to a 10 cGy Cs-137 gamma-ray dose (delivered over 48 h) equally attenuated residual stressful effects in the progeny of bystander cells that were co-cultured with cells irradiated with 200 cGy and in their respective control. The levels of the stress responsive proteins, phospho-p53 (serine 15) and p21Waf1 were decreased in the progeny of control and affected bystander cells following the chronic exposure to 10 cGy from gamma-rays. k) Of particular significance, two separate experiments, performed during NSRL 8A and 8B, have now indicated that the progeny of bystander cells co-cultured with energetic iron ion exposed cells harbor persistently high level of oxidized proteins. l) A preliminary experiment has shown that exposure of the progeny of bystander cells that were co-cultured with cells exposed to 20 cGy from 1 GeV protons expressed an adaptive response when challenged with a dose of 50 cGy from 1 GeV/amu iron ions. IV. Development of a theoretical model (related to Specific Aims 1-3): To better understand the cellular mechanisms associated with bystander effects, we are collaborating with Professor Jean-Paul Jay-Gerin (University of Sherbrooke, Canada) to develop a theoretical model that relies on knowledge of water radiolysis as a prerequisite to understanding the chemistry that takes place in a cellular environment traversed by ionizing particle tracks of different LET and of the resulting cellular changes. It is hoped that this model, which focuses on the production, concentration and diffusion of water radiolysis products will contribute to better understanding of the cellular mechanisms associated with expression of ionizing radiation-induced bystander effects.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Abstracts for Journals and Proceedings	Azzam E. "Ionizing Radiation-Induced Bystander Effects: Mediating Factors and Late Effects." 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-24, 2008. Abstract Book, 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-24, 2008. , Sep-2008

Abstracts for Journals and Proceedings	Buonanno M, Autsavapromporn N, Yang Z, Pandey BN, de Toledo SM, Howell RW, Azzam E. "Late Effects in Irradiated Normal Human Fibroblasts are Dependent on Radiation Quality." 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-24, 2008. Abstract Book, 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-24, 2008. , Sep-2008
Abstracts for Journals and Proceedings	Autsavapromporn N, de Toledo SM, Yang Z, Buonanno M, Zhang J, Harris AL, Jay-Gerin JP, Azzam E. "Participation of gap-junction communication in cell killing and repair of potentially lethal damage in human cell cultures exposed to low- or high LET ionizing radiations." 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-24, 2008. Abstract Book, 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 21-24, 2008. , Sep-2008
Abstracts for Journals and Proceedings	Autsavapromporn N, de Toledo SM, Buonanno M, Yang Z, Harris AL, Jay-Gerin JP, Azzam E. "Modulation of human cell responses to space radiation by gap-junction communication." 37th COSPAR Scientific Assembly, Montreal, Quebec, Canada, July 13-20, 2008. Abstract Book, 37th COSPAR Scientific Assembly, Montreal, Quebec, Canada, July 13-20, 2008. , Jul-2008
Abstracts for Journals and Proceedings	de Toledo SM, Li M, Yang Z, Autsavapromporn N, Buonanno M, Howell RW, Pain D, Harris A, Azzam E. "Bystander Effects in Human Cell Responses to Space Radiation Exposures: The Role of Gap-Junction Communication and Oxidative Metabolism." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30 – July 2, 2008. Abstract Book, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30 – July 2, 2008. , Jul-2008
Abstracts for Journals and Proceedings	Autsavapromporn N, de Toledo SM, Buonanno M, Yang Z, Zhang J, Gonon G, Messungnoen J, Harris AL, Jay-Gerin JP, Azzam E. "The Role of Gap-Junction Intercellular Communication in the Biological Effects of Space Radiation." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30 – July 2, 2008. Abstract Book, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30 – July 2, 2008. , Jul-2008
Abstracts for Journals and Proceedings	Buonanno M, Autsavapromporn N, Yang Z, de Toledo SM, Howell RW, Azzam E. "Long-Term Effects in Irradiated Normal Human Cells are Dependent on Radiation Quality: The Contribution of Bystander Effects and Adaptive Responses." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30-July 2, 2008. Abstract Book, 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, Pennsylvania, June 30-July 2, 2008. , Jul-2008
Abstracts for Journals and Proceedings	Azzam EI. "Ionizing Radiation-Induced Bystander Effects: Mediating Factors and Late Effects." American Statistical Conference on Radiation and Health: "New Developments and Future Directions in Radiation Research", Vail, Colorado, June 15-18, 2008. Abstract Book, American Statistical Conference on Radiation and Health: "New Developments and Future Directions in Radiation Research", Vail, Colorado, June 15-18, 2008. , Jun-2008
Abstracts for Journals and Proceedings	Azzam EI. "Ionizing Radiation-Induced Bystander Effects: Mediating Factors and Late Effects." Annual Meeting of the Association for Radiation Research, University of Manchester, Manchester, UK, March 18-20, 2008. Abstract Book, Annual Meeting of the Association for Radiation Research, University of Manchester, Manchester, UK, March 18-20, 2008. , Mar-2008
Abstracts for Journals and Proceedings	de Toledo SM, Autsavapromporn N, Buonanno M, Yang Z, Li M, Howell RW, Pain D, Harris AL, Azzam E. "The Role of Gap-Junction Communication and Oxidative Metabolism in the Biological Effects of Space Radiation." NASA Human Research Program Investigators' Workshop, League City, TX, February 4-6, 2008. Abstract Book, NASA Human Research Program Investigators' Workshop, League City, TX, February 4-6, 2008. , Feb-2008
Abstracts for Journals and Proceedings	Li M, Boominathan A, de Toledo SM, Pain D, Azzam E. "Aconitase Activity: A Sensitive Metabolic Marker of Ionizing Radiation Exposure." DOE Low Dose Radiation Research Program, Workshop VII, Washington, D.C., January 21-23, 2008. Abstract Book, DOE Low Dose Radiation Research Program, Workshop VII, Washington, D.C., January 21-23, 2008. , Jan-2008
Abstracts for Journals and Proceedings	Buonanno M, Autsavapromporn N, Yang Z, de Toledo SM, Howell RW, Azzam E. "Long-term Effects in Irradiated Normal Human Cells are Dependent on Radiation Dose and Quality." DOE Low Dose Radiation Research Program, Workshop VII, Washington, D.C., January 21-23, 2008. Abstract Book, DOE Low Dose Radiation Research Program, Workshop VII, Washington, D.C., January 21-23, 2008. , Jan-2008
Abstracts for Journals and Proceedings	Azzam E. "Ionizing-radiation induced bystander effects: mediating mechanisms and impact on cancer risk." 2008 American Statistical Association Conference on Radiation and Health, Vail, Colorado, June 15-18, 2008. 2008 American Statistical Association Conference on Radiation and Health, Vail, Colorado, June 15-18, 2008. , Jun-2008
Articles in Peer-reviewed Journals	Azzam E. "Ionizing-radiation induced bystander effects: mediating mechanisms and impact on cancer risk." Radiation Research, in press, expected publication February 2009. , Nov-2008
Articles in Peer-reviewed Journals	Gaillard S, Pusset D, de Toledo SM, Azzam EI, Fromm M. "Distance distribution of bystander effects in alpha-particle irradiated cell populations using a CR-39-based culture dish." Radiation Measurements 2008 Aug;43 Suppl 1:S34-40. http://dx.doi.org/10.1016/j.radmeas.2008.03.063 , Aug-2008
Articles in Peer-reviewed Journals	Azzam EI, de Toledo SM, Pandey PN, Venkatachalam P. "Mechanisms underlying the expression and propagation of low dose/low fluence ionizing radiation effects." International Journal of Low Radiation, 2007 Jul 11;4(1): 61-8. http://dx.doi.org/10.1504/IJLR.2007.014490 , Jul-2007

Papers from Meeting Proceedings	Azzam E, Little JB. "The Impact of Bystander Effects and Adaptive Responses in the Health Risks of Low Dose Ionizing-Radiation: the Modulating Effect of Linear Energy Transfer." 12th International Congress of the International Radiation Protection Association, Buenos Aires, Argentina, October 19-24, 2008. Proceedings, 12th International Congress of the International Radiation Protection Association, Buenos Aires, Argentina, October 19-24, 2008. In press, November 2008. , Nov-2008